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Reskilling for Resilience: Preparing the Energy Workforce for Disruption

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Abstract

This paper explores how targeted workforce capabilities contribute to project resilience in upstream oil and gas, with a focus on integrated operations across drilling, completions, and stimulation. Drawing on responses from 170 experienced project professionals, the study examines three core capabilities, Creative Problem-Solving Capacity (CPSC), Project Management Maturity (PMMM), and Data Analytics Capability (DACP), and their influence on a project team's ability to absorb disruption and adapt under pressure, in a project environment, or known as Project Resilience (PRES). Using structural equation modeling, the results confirm that all three capabilities significantly enable resilience, with creative problem-solving emerging as the most influential. Project management maturity, while important, showed a smaller effect size, reflecting the high baseline of structured governance typically present in oil and gas projects. Data analytics capability also proved critical, particularly in transforming operational data into real-time decision support. Multi-group analysis showed that these relationships held consistently across MENA and non-MENA regions, reinforcing their global relevance. Beyond the statistical findings, the study emphasizes the urgency of workforce reskilling as a strategic imperative. In project environments defined by volatility and complexity, resilience cannot be engineered solely through systems, it must be developed through people. This includes rethinking how organizations cultivate cognitive flexibility, cross-functional awareness, and data fluency within their teams. Recommendations center on embedding resilience-enabling capabilities into structured talent strategies through a blend of training, rotational assignments, and experience-based development. Competency models should evolve to integrate assessment and growth pathways for roles beyond project managers, including drilling engineers, well site supervisors, supply chain managers, and technical leads. The findings position Human Resources and competency leaders as critical partners in project delivery, responsible for aligning workforce development with project complexity and disruption readiness. The paper concludes by proposing future research directions, including sectoral comparisons with downstream operations, deeper regional analyses, and expanded exploration of emerging workforce capabilities that can shape organizational adaptability in high-reliability industries.

Introduction

The upstream oil and gas industry continues to operate in an environment marked by complexity, volatility, and disruption. Integrated projects, those involving tightly sequenced scopes across drilling,

completions, stimulation, and well testing, are particularly vulnerable to external shocks and internal misalignments. During the global pandemic of 2020–2022, organizations managing such projects were forced to rapidly adapt their operating models, workforce configurations, and decision-making processes. While some projects buckled under pressure, others displayed remarkable adaptability, absorbing the disruption, modifying their strategies, and resuming operations with minimal delay. This study focuses on understanding the underlying capabilities that contributed to such project resilience.

The concept of project resilience has gained traction in both academic and industry circles, evolving beyond its roots in risk management to reflect an organization's dynamic capacity to respond to unanticipated events. According to [Rahi and Bourgault \(2022\)](#), project resilience is not a static attribute but a dual-dimensional capability consisting of awareness (the ability to detect and anticipate disruption) and adaptive capacity (the ability to adjust project strategies and workflows in response). Their scale, originally validated in the IT sector, is adopted and adapted here to fit the realities of oil and gas project environments, specifically, upstream integrated projects characterized by high capital intensity and tightly coupled activities.

As part of a broader effort to identify the capabilities most relevant for resilient project execution, McKinsey & Company (2021) surveyed 700 global executives during the pandemic. The consensus was clear: reskilling and upskilling the workforce was a top strategic priority. Of the 25 future-ready competencies identified, three cognitive capabilities stood out: creative problem-solving, project management maturity, and data analytics capability. Each represents a distinct yet interrelated area of human and organizational capital that enables teams to respond effectively to uncertainty. Rather than examining resilience as a byproduct of external conditions, this paper focuses on how these internal capabilities contribute to resilience at the project level.

Talent Management and Resilience

In highly technical sectors such as oil and gas, talent management has historically emphasized compliance, domain expertise, and execution discipline. However, the turbulence of recent years has prompted a shift toward more adaptive and future-oriented talent strategies. Scholars such as [Lengnick-Hall et al. \(2011\)](#) argue that workforce agility, enabled by continuous learning, flexible role design, and decentralized decision-making, is foundational to resilience. This perspective aligns with the dynamic capabilities view (DCV), which positions workforce adaptability as a core enabler of organizational renewal in changing environments ([Teece, 1997](#)).

From this lens, talent is not merely an operational input but a strategic resource that must be cultivated intentionally. This study integrates the role of talent by examining how workforce-based capabilities, namely Creative Problem-Solving Capacity (CPSC), Data Analytics Capability (DACP), and Project Management Maturity (PMMM), contribute to Project Resilience (PRES). These capabilities are not only shaped by individual skill but also by organizational support structures, including training programs, knowledge-sharing platforms, and governance frameworks. By embedding talent within the resilience discussion, this paper offers a more holistic view of capability building in project environments.

Creative Problem-Solving Capacity

Creative problem-solving capacity (CPSC) refers to a team's ability to frame problems in novel ways, generate alternative solutions, and navigate ambiguity using intuition and logic. [Reiter-Palmon and Illies \(2004\)](#) emphasize that creativity in project settings is not a luxury but a necessity, particularly when predefined workflows are disrupted and standard operating procedures no longer apply. [Carmeli et al. \(2013\)](#) expand this view by demonstrating how creative behavior at the team level fosters cognitive flexibility, allowing project actors to respond fluidly to evolving constraints.

In this study, the measurement of CPSC follows [Carmeli's \(2021\)](#) work on team creativity in dynamic environments. It operationalizes problem-solving as a collective, strategic activity shaped

by trust, information sharing, and the freedom to challenge assumptions. Creative problem-solving becomes especially salient in upstream oil and gas projects, where subsurface uncertainties, vendor interdependencies, and evolving safety requirements often collide in real-time decision-making.

Project Management Maturity

The maturity of project management processes has long been associated with efficiency, standardization, and risk control. However, recent perspectives recast Project Management Maturity (PMMM) as an enabler of adaptability, providing structure without rigidity. De Bruin and Rosemann (2010) argue that higher maturity allows for faster coordination, more reliable scenario planning, and quicker mobilization of recovery plans. Dijkman et al. (2015) similarly note that process maturity enhances situational awareness, a core component of project resilience. In the context of oil and gas engineering, Ruzhnikov (2024) found that greater maturity in Well Engineering Management Systems (WEMS) was positively correlated with the adaptive capability of engineering teams. This company-level study underscores how maturity not only ensures operational discipline but also enables flexible responses to disruption.

In this study, PMMM is measured using a modified version of the Business Process Management Maturity (BPMM) model, adapted to suit the operational realities of integrated projects in the energy sector. The model assesses both the structural consistency of PM practices (e.g., governance, knowledge management, lessons learned) and their adaptive responsiveness (e.g., change agility, role clarity during crisis response). Maturity is not viewed as mere procedural discipline, but as a dynamic asset that supports rapid adaptation under pressure. This perspective continues the work of Prasetyo et al (2024) which builds on Sincora (2022) where adaptability should be the way to measure maturity under stress, over rigidity of the individual project management processes.

Data Analytics Capability

Data Analytics Capability (DACP) refers to the organization's ability to acquire, process, interpret, and act upon data to guide project decisions. Dubey et al. (2019) highlight DACP as a composite capability comprising technical infrastructure, analytical competence, and organizational routines that translate data into insight. Srinivasan and Swink (2018) further argue that DACP facilitates decision-making flexibility, a critical enabler of project-level responsiveness in the context of supply chain.

Measuring Project Resilience

Prior research has examined organizational resilience through various frameworks, typically capturing the capacity to manage immediate challenges and adapt to evolving conditions. Stephenson (2010) measured resilience using planning and adaptive capacity dimensions across thirteen indicators in New Zealand-based organizations. McManus (2008) introduced the Relative Overall Resilience (ROR) model with a focus on situation awareness, keystone vulnerabilities, and adaptive capacity, assessed through fifteen indicators. Building on these, Lee et al. (2013) retained the core dimensions and validated them in a broader context. Subsequent studies, such as Yang and Cheng (2020), applied this framework to Chinese construction megaprojects, refining it by including indicators like digital technology use, redundancy, and public participation.

Zhang et al. (2023) proposed a three-phase model: Plan, Cope, and Adapt, integrating the capability-based and processual perspectives of resilience. Their framework views project resilience as a dynamic attribute of temporary organizations, emphasizing anticipation, response, and learning. To support resilience, they recommend tools like scenario planning, team simulations, and structured contingency playbooks. This study adopts the two-dimensional model of Rahi and Borgault (2022), continued by Prasetyo et al (2025) which balances conceptual robustness with ease of survey implementation. Their framework defines "awareness" as proactive readiness and "adaptive capacity" as the ability to respond and recover during and

after disruption. Compared to other models, it offers clarity in operationalization while capturing essential features of resilience.

Methodology

This study employed a quantitative, cross-sectional research design to examine the role of three workforce-related capabilities, Creative Problem-Solving Capacity (CPSC), Project Management Maturity (PMMM), and Data Analytics Capability (DACP), in shaping Project Resilience (PRES) within the upstream oil and gas sector. The data were collected through a structured survey instrument distributed to project managers and senior professionals involved in integrated projects across drilling, completions, and stimulation activities. The survey was conducted over a four-month period. To ensure relevance and consistency, participants were required to meet three pre-conditions: (1) a minimum of five years of experience in project execution roles, (2) job titles indicating accountability or leadership, and (3) direct involvement in project delivery during the global disruption of 2020–2022. This ensured that all respondents had firsthand exposure to navigating disruption. Respondents come from multiple entities (rig contractors, service companies and operators).

Using G*Power software for power analysis, the minimum sample size required was calculated to be 119. This was determined using a priori power analysis for linear multiple regression with a fixed model and three predictors, targeting a power level of 0.95 and a medium effect size. This provides statistical justification for the adequacy of the sample size used in this study. Constructs were measured using indicators from prior validated studies: Carmeli (2021) for CPSC, Dijkman (2015) for PMMM, Dubey et al. (2019) and Akter et al. (2016) for DACP, and Rahi and Bourgault (2022) for Project Resilience (PRES), using five-point Likert scales. The questionnaire was distributed electronically through convenience and snowball sampling.

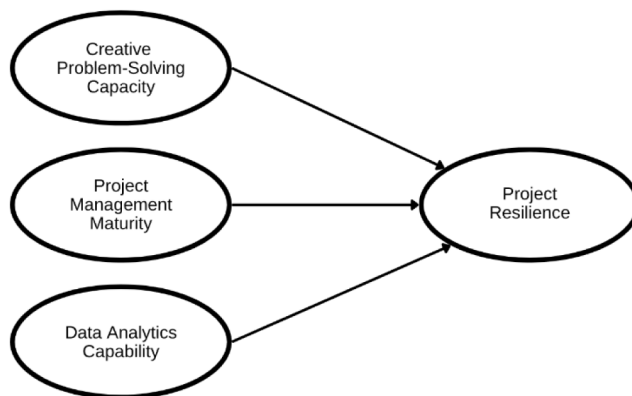


Figure 1—Research Model

Model evaluation followed Hair et al. (2019), assessing both measurement and structural models. A summary of construct indicators and their sources is provided in Appendix 1 for reference. While technical rigor was maintained, the analysis was designed to remain accessible and applicable to industry professionals.

To explore regional differences, a Multi-Group Analysis (MGA) was conducted, comparing responses between MENA and non-MENA regions. Beyond satisfying statistical thresholds, this segmentation reflects distinct industry contexts. In the MENA region, integrated projects are increasingly common, supply chains depend more on global vendors, and national oil companies dominate, sometimes as single operators. In contrast, non-MENA regions feature more varied operator types, stronger local supply networks, and diverse contracting models. These structural differences provide meaningful insight into how resilience capabilities may function differently across regions.

In parallel with the quantitative analysis, a qualitative review panel was established to contextualize the findings. The panel comprises four subject matter experts in project management and one specialist in talent development and competency management. Their role is to provide interpretive insight, assess practical implications, and co-develop the recommendations and application framework based on the study's results.

While the core of this study is quantitative, the inclusion of expert insights through focused discussions and interviews enriches the interpretation of findings and grounds the recommendations in practical experience.

Results and Discussion

The survey collected a total of 170 valid responses over a four-month period. In terms of professional background, 43% of respondents held project management titles, 27% were from operations functions, and the remaining 30% represented disciplines including engineering, quality, and digital enablement. Regarding seniority, the majority (66%) had more than 10 years of working experience, while 21% had 6–10 years, and the rest had between 5–6 years. Geographically, the sample was well-distributed, with 41% of respondents from the Middle East, 28% from Asia-Pacific, 18% from Europe, and the remainder from North and South America.

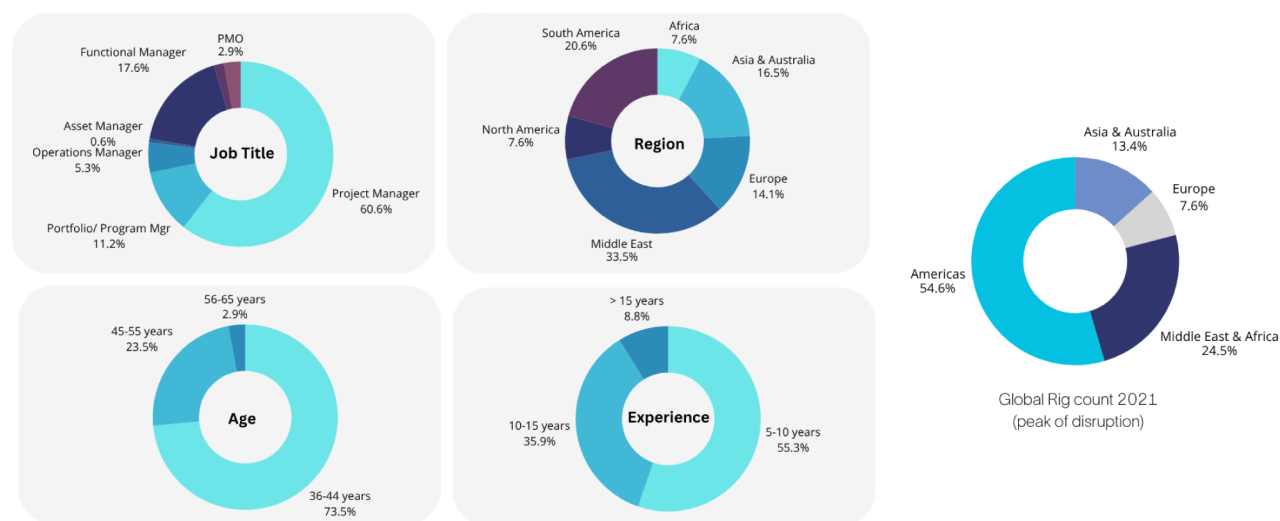


Figure 2—Respondents Profile

Measurement Model Evaluation

The measurement model was evaluated based on standard Partial Least Squares – Structural Equation Modeling (PLS-SEM) reliability and validity criteria. Composite reliability (CR) values for all constructs exceeded the 0.70 threshold, indicating strong internal consistency. Average Variance Extracted (AVE) values were above 0.50, supporting convergent validity. Discriminant validity was established using the Heterotrait-Monotrait (HTMT) ratio, with all construct pairs falling below the 0.90 cut-off. The model contains a multi-order construct and therefore the evaluation will be done following Hair's (2019) two-step evaluation which is to evaluate the measurement model at every order and followed by a structural model evaluation.

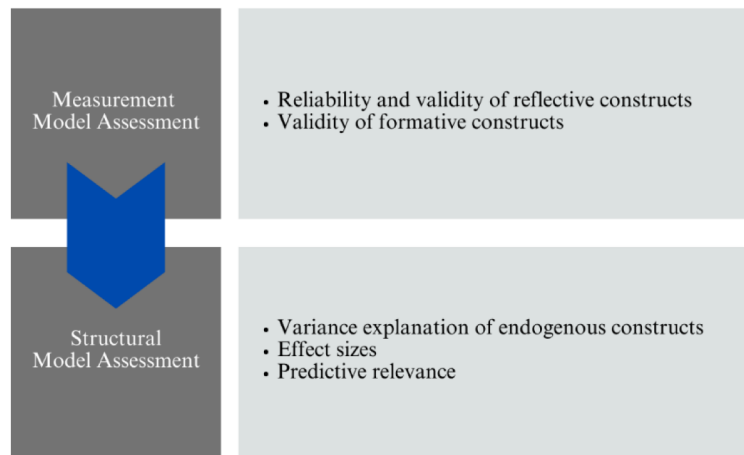


Figure 3—Two-Step Process of PLS Path Model Assessment

During first-order measurement model refinement, a few from Project Resilience (PRES) were removed due to low outer loadings. This is done for statistical requirements and does not affect the conceptual integrity of the measurement. At the second order evaluation, AWRN (Awareness) was retained despite marginal values to preserve theoretical structure of Project Resilience, consistent with [Rahi and Bourgault's \(2022\)](#) validated dimensions. Retaining indicators that marginally fail the AVE requirement, due to theoretical/ conceptual integrity is also in line with [Cheung \(2024\)](#). Likewise, for Project Management Maturity (PMMM), Level 0 ("initial") was already excluded from the model and questionnaire, as an absence of structured process is rarely observed in the oil and gas sector, where formal governance frameworks are directly or indirectly mandated by oil and gas operators during tendering and execution. Selected marginal indicators for CPSC and PMMM were retained based on their conceptual relevance to integrated upstream project environments. [Table 1](#) details the second-order measurement model evaluation results, and [Table 2](#) details the final third-order measurement model evaluation results where all the construct dimensions meet the statistical quality requirement.

Table 1—Second-Order Measurement Model Evaluation

Second-Order Constructs	Composite Reliability (ρ_c)	Average Variance Extracted (AVE)
Accessibility and Mobilization of Resources (MOBR)	0.755	0.51
Adaptive and Responsible Decision-Making (DCMK)	0.897	0.745
Alertness of Performance Deviation (ALPD)	0.817	0.601
Creative Problem-Solving Capacity (CPSC)	0.821	0.535
Data Analytics Capability (DACP)	0.824	0.54
Effectiveness of Communications and Relationships (COMM)	0.778	0.541
Efficiency of External Resources (EXRS)	0.886	0.721
Maturity Level 2 Managed (MNGD)	0.838	0.633
Maturity Level 3 Standardized (STDD)	0.755	0.511
Maturity Level 4 Predictable (PDCT)	0.783	0.644
Maturity Level 5 Innovating (INVT)	0.838	0.634
Responsiveness of Team Members (RSPV)	0.787	0.555
Sensitivity to Organizational Changes (SOCH)	0.816	0.598
Use of Skills and Expertise (SKLE)	0.733	0.487

Table 2—Third-Order Measurement Model Evaluation

Third-Order Constructs	Composite Reliability (ρ_c)	Average Variance Extracted (AVE)
Creative Problem-Solving Capacity (CPSC)	0.808	0.514
Data Analytics Capability (DACP)	0.822	0.537
Project Management Maturity (PMMM)	0.836	0.561

Structural Model Results (Path Coefficients)

The structural model results confirmed that all three workforce capabilities had significant positive effects on Project Resilience. Creative Problem-Solving Capacity (CPSC) showed the strongest effect ($\beta = 0.317$, $t = 3.870$, $p = 0.000$), followed by Data Analytics Capability (DACP) ($\beta = 0.243$, $t = 2.790$, $p = 0.005$), and Project Management Maturity (PMMM) ($\beta = 0.214$, $t = 2.548$, $p < 0.001$). The R^2 value for Project Resilience was 0.42, indicating that the model explains 42% of the variance in resilience outcomes among project teams. All variance inflation factor (VIF) values were well below 3.3, confirming the absence of multicollinearity. Table 3 details the results of the structural model evaluation results.

Table 3—Structural Model Evaluation (Path Coefficients)

Path (Hypotheses)	Path Coefficient(β)	T statistics	P value
Creative Problem-Solving Capacity (CPSC) -> Project Resilience (PRES)	0.317	3.87	0.000
Data Analytics Capability (DACP) -> Project Resilience (PRES)	0.243	2.79	0.005
Project Management Maturity (PMMM) -> Project Resilience (PRES)	0.214	2.548	0.011

In terms of effect size (f^2), PMMM demonstrated a small but statistically meaningful effect on Project Resilience ($f^2 = 0.03$). This indicates that while project management maturity is not the dominant driver, it still plays an enabling role by providing governance consistency, scenario readiness, and structured integration across functions. In upstream environments, even modest maturity gains can support better decision flow, stakeholder coordination, and risk mitigation under disruptive conditions. The relatively small effect size for PMMM may be partially attributed to the high overall maturity levels observed among survey respondents. As noted by the expert panel, many tenders in the oil and gas sector explicitly require evidence of structured project management experience, including documentation of integrated project delivery, governance frameworks, and lessons learned. This baseline of maturity likely minimizes variability across the sample, making PMMM a norm rather than differentiating factor. Only a handful of respondents reflected ad-hoc or reactive project environments, often characterized by informal coordination and tacit knowledge, suggesting that mature practices are already embedded in most upstream operations.

These results support the theoretical proposition that individual and organizational capabilities centered on creative problem-solving, structured project management, and data-driven decision-making directly enhance a project team's capacity to absorb and adapt to disruption.

Interpretation of Key Relationships

The strongest predictor of project resilience was Creative Problem-Solving Capacity (CPSC), reflecting the importance of adaptability and cognitive flexibility during crisis situations. This finding reinforces Carmeli's (2021) assertion that creative teams demonstrate higher levels of initiative and improvisation under pressure. In upstream project settings, where teams routinely face evolving operational parameters and subsurface uncertainties, the ability to frame problems innovatively and develop context-sensitive solutions becomes a critical determinant of sustained performance.

Project Management Maturity (PMMM) also had a statistically significant impact on resilience. This supports prior literature (De Bruin & Rosemann, 2010; Dijkman et al., 2015) positioning maturity as more than just procedural discipline, it reflects an organization's capacity for structured response and organizational learning. In environments with high interdependence between planning and execution, such as integrated well delivery, mature governance systems enable timely decision-making, scenario planning, and mitigation protocols. The result affirms that maturity does not hinder adaptability but rather enables a stable foundation for it. This is particularly relevant given that the metric used to measure PMMM in this study is based on the Business Process Management Maturity (BPMM) model, which includes adaptability as one of its core pillars. While mature processes may appear rigid, the presence of clearly defined management of change protocols allows those same processes to be rapidly adjusted during periods of disruption. In this way, maturity supports strategic responsiveness by minimizing confusion and enabling faster adaptation under pressure. It transforms adaptability from an ad-hoc reaction into a repeatable organizational capability.

Finally, Data Analytics Capability (DACP) emerged as a meaningful contributor to resilience. Drawing from Dubey et al. (2019) and Akter et al. (2016), this construct encompasses not only technical infrastructure but also workforce readiness in interpreting and applying data. In the context of upstream projects, real-time data from rig operations, supply chains, and HSE events must be processed quickly to support agile decision-making. The result suggests that analytics maturity can transform data into foresight, improving teams' capacity to anticipate, absorb, and act in volatile situations.

Regional Insights into Resilience Capabilities (Multi-Group Analysis)

A Multi-Group Analysis (MGA) was conducted to compare MENA and non-MENA regions, testing whether the relationships between CPSC, PMMM, DACP, and Project Resilience varied by geography. Splitting the category to MENA and non-MENA regions is statistically grounded considering the distribution of respondents (70 from MENA and 100 from non-MENA), but also because there are differences in the project environment. In MENA, the projects are more integrated (bundling of multiple services with incentives, full lumpsum turnkey or joint ventures), many MENA countries operate in a single or limited operator environment (usually state owned), with strict import rules and heavy reliance on a global supply chain network. Projects in non-MENA countries are less integrated (bundling is done for coordination purpose), with more independent operators alongside state-owned operators working, and a more mature local supply chain network in combination with global networks for complex technology.

The results in Table 4 showed that all three constructs had statistically consistent effects across both groups, with no significant differences in path coefficients. This suggests that the proposed resilience-enabling capabilities are robust and applicable across diverse project environments, regardless of geography.

Table 4—Multi-Group Analysis (MGA) for MENA and Non-MENA Regions

Path (Hypotheses)	Difference (MENA – Non-MENA)	2-Tailed p Value (MENA vs Non-MENA)
Creative Problem-Solving Capacity (CPSC) -> Project Resilience (PRES)	-0.058	0.726
Data Analytics Capability (DACP) -> Project Resilience (PRES)	0.204	0.229
Project Management Maturity (PMMM) -> Project Resilience (PRES)	0.018	0.928

Expert panel feedback supported these findings, noting that while operational nuances exist, such as more centralized control in MENA projects or more agile supply networks in non-MENA, the foundational value of Creative Problem-Solving Capacity (CPSC), Project Management Maturity (PMMM), and Data Analytics Capability (DACP) remain universal. This indicates that organizations seeking to enhance project

resilience can confidently apply these capability areas globally, while tailoring delivery to their local contexts.

Workforce Reskilling for Resilience: Bridging Insight and Action

The COVID-19 pandemic was more than a disruption; it was a capability stress test. A 2021 McKinsey survey of over 700 global executives found a striking consensus: the most urgent workforce priority was reskilling (McKinsey, 2021). Among the top future-proof capabilities identified were problem-solving, data literacy, and project management. This study empirically reinforces that call. The statistically significant contribution of Creative Problem-Solving Capacity (CPSC), Project Management Maturity (PMMM), and Data Analytics Capability (DACP) to project resilience confirms their strategic relevance, not as abstract competencies, but as resilience-enabling tools that teams rely on during extreme volatility.

From Capability Gaps to Development Pathways

From a talent management perspective, these findings offer direction. Resilience cannot be outsourced; it must be built into the workforce. For CPSC, this means creating experiential learning environments where teams are routinely exposed to complex, ambiguous challenges, in a psychologically safe and growth-oriented culture. Exposure to a wide variety of real project problems, especially those encountered under pressure, shapes an individual's ability to respond creatively and constructively. One way to facilitate this development is through short-term assignments, rotation programs, and deliberate team composition strategies. Projects benefit from a mix of local and global knowledge, technical and soft skill backgrounds, and varying levels of seniority. When problem-solving capability is seen as a product of experience diversity, organizations can proactively design teams and assignments to cultivate resilience. Core competencies in this domain include scenario framing, ideation under constraint, facilitation of divergent thinking, and collaborative decision-making under ambiguity. Structured problem-solving frameworks such as TRIZ (a systematic innovation methodology originating from engineering problem patterns) or design thinking (a user-centered, iterative approach to complex problem-solving) may complement this, but are most effective when grounded in direct, varied project exposure.

PMMM, on the other hand, requires more than project management certification; it demands an ecosystem of shared practices embedded in day-to-day project delivery. The direction is toward a competency model that evolves to assess not just technical execution, but practical leadership in integrated environments, governance fluency, and stakeholder coordination. Competencies such as integrated change control, risk-based scenario planning, and integrated team management reflect this maturity in action. When change management processes are clearly defined and activated during crises, even structured systems become highly responsive.

For DACP, the journey should start by involving team members who understand the project's needs and enabling them to work closely with data scientists or even upskilling them to define solutions that support decision-making based on accurate, relevant, and context-specific data. Cultivating data curiosity, pattern recognition, and the ability to interpret operational analytics should become a core skill for all project team members. Competencies such as data storytelling, root cause analytics, and visual interpretation of operational metrics are increasingly critical. Training programs and development pathways should emphasize the practical, contextual use of data to support everyday decisions, ensuring that data is accessible and presented in a shared language that the entire project team can understand and apply. The use of digital tools for data visualization, analysis, and interpretation should be promoted as a standard capability across teams, supported by broader organizational commitment.

The Role of Talent and Competency Leaders

Competency leads and L&D professionals must rethink how capability frameworks are built and tracked. In integration projects, particularly in upstream oil and gas, require cross-functional awareness and resilience

as a core outcome, not just technical delivery. Reskilling programs must align with project realities: fast-paced turnarounds, contractor-based ecosystems, and culturally diverse teams. This requires building modular learning paths and directly linking development plans to project complexity profiles.

While project managers were the primary respondents in this study, the findings have broader implications. Key roles such as drilling engineers, well site supervisors, supply chain managers, and drilling managers are essential to upstream operations and equally stand to benefit from targeted capability development. Organizations should therefore explore how to elevate team-wide competencies through integrated development efforts that support both individual skill growth and collaborative team performance.

Competency development in this context should be anchored in three foundational elements: training, experience, and assessment. To build resilience-oriented capabilities, organizations must ensure that training is standardized, experience is systematically tracked, and qualified assessors are in place. While building such a model takes time, resources, and attention, it is essential for developing a workforce equipped with efficiency, agility, and, above all, resilience.

This is not a one-time initiative; competency systems must remain dynamic, continuously evolving to keep pace with the growing complexity of integration projects, technological advancements, and changing operational demands.

HR as a Strategic Partner in Project Delivery

Human Resource teams have a critical opportunity to reposition themselves from support function to strategic enabler. The evidence is clear: organizations that proactively develop CPSC, PMMM, and DACP within their workforce improve not only adaptability but project continuity and delivery quality. In high-reliability sectors like oil and gas, talent development is not a cost center, it is a resilience multiplier. Industry literature supports this shift: organizations that embed dynamic capabilities within workforce systems show greater long-term adaptability (Teece, 1997).

Workforce transformation must now be understood as project risk mitigation. The pathway forward lies in cross-functional collaboration between HR, PMOs, and business line leaders to institutionalize resilience, not only through systems, but through people.

Conclusion and Recommendations

This study highlights the pivotal role of three workforce capabilities, Creative Problem-Solving Capacity (CPSC), Project Management Maturity (PMMM), and Data Analytics Capability (DACP), in enabling project resilience across upstream oil and gas environments. Amid growing volatility and complexity, these capabilities serve as critical levers that empower teams to absorb disruption, adapt under pressure, and continue delivering value. The empirical evidence confirms that their impact is consistent across regional contexts, reinforcing their universal relevance for global project organizations.

For practitioners, the results signal the importance of embedding these capabilities not only in individual roles but across project teams and functions. Workforce development strategies should evolve to ensure exposure to varied problem contexts, structured competency models anchored in experience and assessment, and a stronger partnership between technical and talent development leaders. The inclusion of HR as a strategic partner in project delivery is no longer optional, it is essential.

Future research can build on these insights by exploring additional resilience-enabling capabilities identified in global surveys, such as collaboration, emotional intelligence, and digital agility. Studies could also extend into downstream project environments, where stakeholder complexity and external volatility may introduce different dynamics. Finally, deeper regional or country-level analyses, beyond the MENA vs. non-MENA split, may uncover contextual nuances that can inform more targeted reskilling and organizational design interventions.

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Appendix 1

Constructs dimensions and indicators

Construct	Dimension	Subdimension	Indicator (Original)	Adopted/ Adapted Indicator
Project Resilience (Rahi and Borgault, 2022) (PRES)	1. Awareness (AWRN)	1. Use of skills and expertise (SKLE)	- I continuously use my expertise to identify project risks - I continuously use my expertise to assess project risks - I work in collaboration with other team members to remain alert to disruptive events	- I continuously use my expertise to identify project risks - I continuously use my expertise to assess project risks The team members work in collaboration with other team members to remain alert to disruptive events
		2. Alertness of performance deviations (APLD)	- Based on project type, project management methods (project plan template, agile, etc.) are employed to capture all project requirements and avoid possible disruption events - Based on project type, risk management methods (risk register, monte carlo, etc.) are used to avoid possible disruptive events - KPIs for the budget are continuously monitored to detect any budget performance deviation throughout the project life cycle - KPIs for the schedule are continuously monitored to detect any schedule performance deviation throughout the project life cycle	- Project Management methods and artifacts (project plan, issue log, dispute logs, incident investigations) are employed to capture all project requirements and avoid possible disruption events - Risk management methods (risk register, risk classification matrix, monte carlo, etc.) are used to avoid possible disruption events - Project financial KPIs (cost, revenue, profitability) and schedule KPIs are continuously monitored to detect any deviation throughout the project Modification is to combine cost and schedule KPIs and providing examples for clarity using common risk tools in upstream oil and gas.
		3. Sensitivity to organizational changes (SOCH)	- Internal organizational changes (new PM process, update in methods, etc.) that may impact the project are clearly communicated to avoid possible disruptive events - When needed, resources within the organization are accessible to help identify project weaknesses - When needed, resources within the organization are accessible to help assess the project weaknesses	- Internal organizational changes (new PM process, update in methods, etc.) that may impact the project are clearly communicated to avoid possible disruptive events - When needed, resources within the organization are accessible to help identify project weaknesses - When needed, resources within the organization are accessible to help assess the project weaknesses
		4. Efficiency of external resources (EXRS)	- Partnership with external parties that have field-specific knowledge are continuously developed - Existing partnerships with external parties are continuously strengthened - Access to external resources through partnerships help to identify project weaknesses - Access to external resources through partnerships is possible to help assess project weaknesses	New partnerships with external parties that have field-specific knowledge are continuously developed - Existing partnerships with external parties are continuously strengthened - Access to external resources through partnerships help to identify and assess project weaknesses Modification is to combine "identify" and "assess", and to clarify new partnerships as part of a proactive attempt to further strengthen resilience.
	2. Adaptive Capacity (ADPT)	1. Accessibility and mobilization of resources (MOBR)	When needed, resources can be mobilized from different departments to face disruptive events	When needed, resources can be mobilized from different departments in the company and

Construct	Dimension	Subdimension	Indicator (Original)	Adopted/ Adapted Indicator
			- When needed, resources can be mobilized through partnerships to face disruptions - Information systems are available to provide quick access to information to solve an issue - Information systems are available to log disruptive events - Departments can provide feedback on possible solutions to face disruptive events through information systems - External partners can provide feedback on possible solutions to face disruptive events through information systems	third-party companies to face disruptive events - Information systems are available to log disruptive events - Involved parties (departments and third parties) can access and provide feedback on possible solutions to face disruptive events through digital tools such as websites, emails or industry-specific software In the context of large projects, there is no necessity to distinguish internal or external resources. The dimension speaks of access and mobilization of resources. Third party partnerships also have been covered in the previous dimension. Information system is clarified with examples.
		2. Responsiveness of team members (RSPV)	- I tend to adopt creative solutions to face disruptive events - I am encouraged to think outside the box to find solutions to face disruptive events - I have the expertise to deal with disruptive events	- I tend to adopt creative solutions to face disruptive events - I am encouraged to think outside the box to find solutions to face disruptive events - I have the expertise to deal with disruptive events
		3. Effectiveness of communications and relationships (COMM)	- Clear explanations of issues and possible solutions are provided to all stakeholders through transparent communications - Stakeholders, including the client, are encouraged to take ownership of issues and possible solutions - Stakeholders are involved through continuous follow-ups (regular meetings, status calls, etc.) to ensure proper management of disruptive events	- Clear explanations of issues and possible solutions are provided to all stakeholders through transparent communications - Stakeholders, including the client, are encouraged to take ownership of issues and possible solutions - Stakeholders are involved through continuous follow-ups (regular meetings, status calls, etc.) to ensure proper management of disruptive events
		4. Adapted and responsible decision-making (DCMK)	- Senior managers plot a course of action to face disruptive events - Senior managers ensure actions are implemented in the right way and at the right time to effectively face disruptive events - Senior managers adapt their strategies to face disruptive events based on the project's importance to the organization	Project sponsors or senior managers plot a course of action to face disruptive events Project sponsors or senior managers ensure actions are implemented in the right way and at the right time to effectively face disruptive events Project sponsors or senior managers adapt their strategies to face disruptive events based on the project's importance to the organization Adding clarity with project sponsors, which is common in project organizations. Project sponsors sit outside of the project team and are usually part of the senior management.
Creative Problem-Solving Capacity (Carmeli, 2013) (CPSC)			- We have the capability to define work problems creatively - We have the skill to creatively articulate work problems - We are able to generate novel ideas to solve work problems - We have the capability to suggest creative solutions to work problems	- Our team effectively identifies work problems using creative approaches such as brainstorming, root cause analysis, scenario planning, etc. - Our team effectively communicates work problems to stakeholders using methods such as visualization, decision trees, etc.

Construct	Dimension	Subdimension	Indicator (Original)	Adopted/ Adapted Indicator
			5. We have the capability to appreciate what ideas are best for solving work problems 6. We have the capability to choose the optimal solution for a specific work problem 7. We have the capability to effectively implement novel ideas chosen to solve a specific work problem 8. We have the capability to implement the chosen creative solution to solve a specific work problem	3. We have the capability to suggest creative solutions to work problems 4. Our team has the capability to evaluate and recognize the most effective ideas for solving work problems, considering factors such as feasibility, impact and practicality 5. We have the capability to effectively implement novel ideas chosen to solve a specific work problem Modification is largely around providing examples for clarity using common techniques used in upstream oil and gas, for problem solving that are incident induced such as (root cause analysis, brainstorming) or proactive planning (scenario planning, feasibility study).
Data Analytics Capability (Dubey et al., 2019; Akter et al., 2016) (DACP)			1. We use advanced tools and analytical techniques (simulation, optimization, regression, etc.) to take decisions 2. We use information extracted from various sources of data to take decisions 3. We use data visualization techniques (dashboard, etc.) to assist users or decision makers in understanding complex information 4. Our dashboards display information which is useful for carrying out the necessary diagnosis 5. We have connected dashboard applications or information to the manager's communication devices 6. The project team utilizes data analytics to help continuously monitor the progress of the project. 7. In our organization there are personnel capable of teaching others about data analytics	1. We use advanced tools and analytical techniques (simulation, optimization, regression, etc.) to take decisions 2. We use information extracted from various sources of data to take decisions 3. We use data visualization techniques (dashboard, etc.) to assist users or decision makers in understanding complex information 4. We have connected dashboard applications or information with the manager's communication devices 5. In our organization there are personnel capable of teaching others about data analytics Main modification is to add the last point regarding availability of personnel to teach data analytics in the organization, and not necessarily within the project itself.
Project Management Maturity (Dijkman, 2015; Sincora 2022) (PMMM)	Initial		1. Formal procedures for the execution of processes do not exist in our project 2. If procedures are defined, they are rarely followed 3. Everyone executes tasks in their own way	Removed – this dimension is a reflective dimension. It is removed because projects in upstream oil and gas during the tendering stage will require participants to demonstrate experience and capability in project management of well construction. It is impossible that projects on this scale will be run without any procedures/ processes in place or relying on ad-hoc solutions.
	Managed (MNGD)		1. At the beginning of the project, we make agreements about which methods and methodology we will use 2. If we make agreements about work methods, they will be documented such that they can be executed in the same way at another time 3. We use planning and management procedures to control our individual projects	1. At the beginning of the project, we make agreements about which methods and methodology we will use 2. If we make agreements about work methods, they will be documented such that they can be executed in the same way at another time 3. We use planning and management procedures to control our individual projects

Construct	Dimension	Subdimension	Indicator (Original)	Adopted/ Adapted Indicator
	Standardized (STDD)		1. Procedures are standardized for the project 2. Work procedures and objectives are well documented in our project 3. Processes are defined such that they will be in the same way by different work groups	1. Procedures are standardized for the project 2. Work procedures and objectives are well documented in our project 3. Processes are defined such that they will be in the same way by different work groups
	Predictable (PDCT)		1. Performance is managed statistically (e.g. by measuring KPIs) to understand performance and control variation 2. Processes and tasks are managed in such a way that they meet agreed-upon performance and quality goals 3. If processes do not perform according to predefined standards, they are corrected to meet the quantitative goals	1. Processes and tasks are managed in such a way that they meet agreed-upon performance and quality goals 2. If processes do not perform according to predefined standards, they are corrected to meet the quantitative goals The first indicator was removed because of duplication with another project resilience question regarding financial and schedule KPIs. In addition to that, all projects in oil and gas will have internal and external KPIs
	Innovating (INVT)		1. Our project understands its critical business issues and areas of concern by using feedback from performance measurements 2. Our project sets quantitative improvement goals to constantly reorganize processes when perceived necessary 3. We constantly pilot new ideas and new technologies to improve our processes	1. Our project understands its critical business issues and areas of concern by using feedback from performance measurements 2. Our project sets quantitative improvement goals to constantly reorganize processes when perceived necessary 3. We constantly pilot new ideas and new technologies to improve our processes